Contact-Aware RL for Robotic Ultrasound: Reward Component Ablation

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Introduction

Project Aim:

- **How do individual reward components (force, distance, and image) contribute to simulated ultrasound scanning performance?**

Reinforcement Learning:

- Trial-and-error learning where the agent takes actions to maximize a cumulative reward. [1]
- The robotic probe learns to autonomously scan for foreign bodies, maintain contact, and minimize tissue deformation.

Ablation Study:

- Four reward configurations are evaluated: all rewards, no image reward, no force reward, and no distance reward.
- Each condition is tested across four noise levels (σ = 0, 0.005, 0.01, 0.015) applied to the ultrasound image input.
- Isolates the contribution of each reward component to training convergence and noise robustness.

Deformation and Contact-Aware Simulation:

- Simulates soft-tissue deformation and contact forces for realistic probe–tissue interaction. [2]

Methods

Configuration File:

```
behaviors:
  CurvilinearProbe:
    trainer_type: sac
    hyperparameters:
      learning_rate: 0.001
      batch_size: 512
      buffer_size: 100000
      buffer_init_steps: 10000
      tau: 0.005
      steps_per_update: 20
      init_entcoef: 0.3
      learning_rate_schedule: constant
    network_settings:
      vis_encode_type: resnet
      normalize: true
      hidden_units: 256
      num_layers: 2
    reward_signals:
      extrinsic:
        gamma: 0.99
        strength: 1.0
    max_steps: 3000000
    summary_freq: 500
    keep_checkpoints: 5
engine_settings:
  time_scale: 1
  target_frame_rate: -1
  capture_frame_rate: 60
  no_graphics: false
```

Figure 1: Reinforcement learning configuration parameters for the ultrasound probe agent, including network architecture, training settings, and reward signal weights used in Unity ML-Agents.

Reward Definitions:

Reward feature ψ_i^r	Weight w_i^r
Step penalty (encourage efficiency)	−0.01
Contact force	
Force within desired range	+5
Force too high	−5
Force too low	−0.01
Tumor detection	
Dark blob (tumor) detected in ultrasound image	+1
No blob detected	−0.01
Probe position	
Change in distance (d) to goal, where $\phi(d) = e^{-2d}$	$+200(\phi_{now} - \phi_{prev})$
Within goal region	+1
Termination conditions	
Task completion (force in range + blob detected + probe in goal region)	+6000
Out-of-bounds penalty	−1000

Figure 2: Reward function components and weights used during training. Task completion requires simultaneous force regulation, tumor detection, and probe positioning, awarding +6000 on success.
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Unity Environment

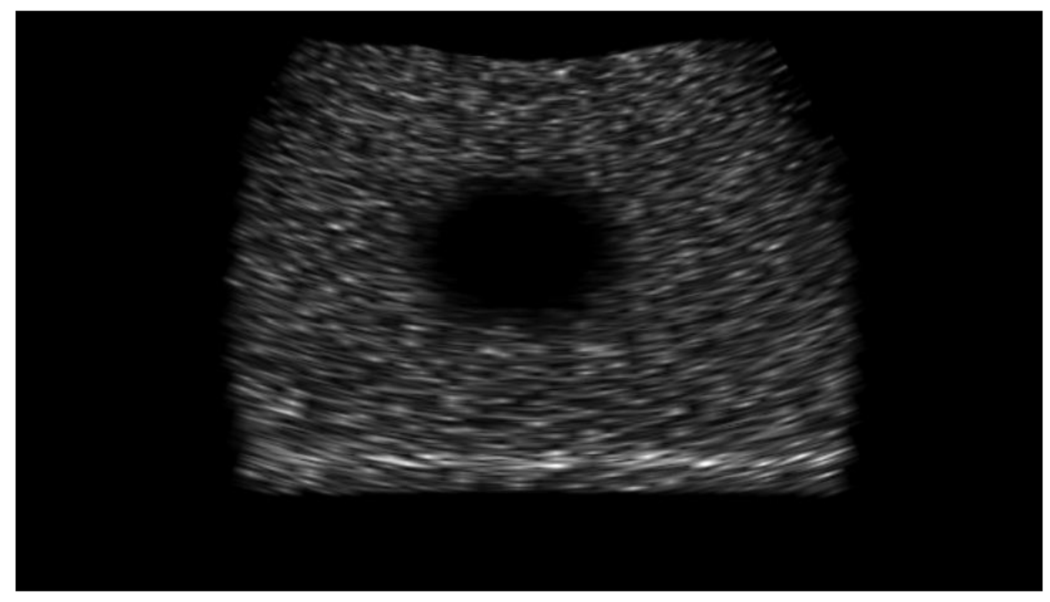

Figure 3: Simulated ultrasound image captured in a low-noise setting (σ = 0), showing a clearly visible dark blob corresponding to the target tumor.

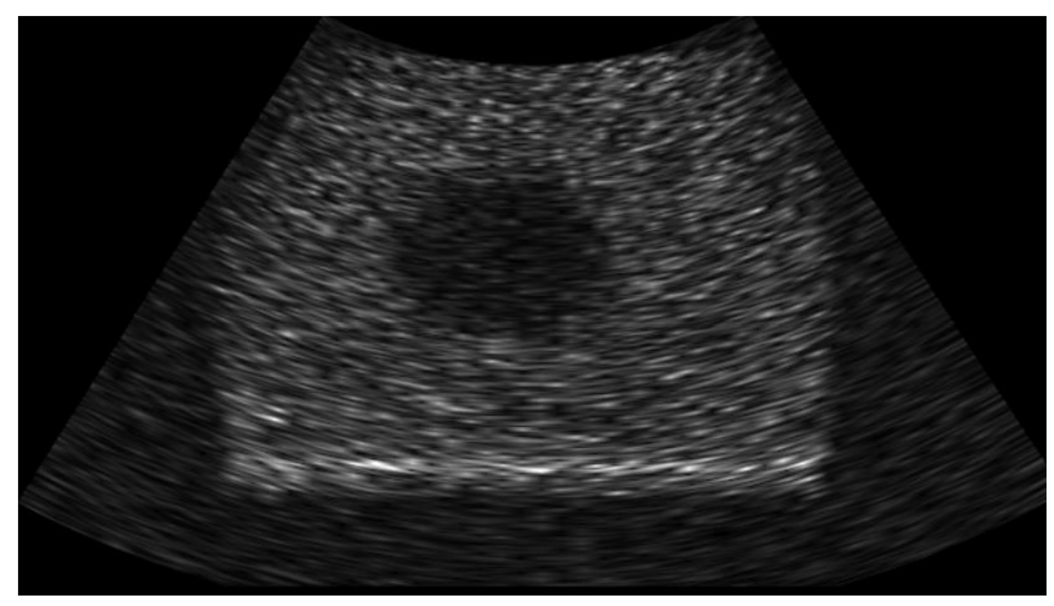

Figure 4: Simulated ultrasound image captured in a high-noise setting (σ = 0.015), showing significant corruption of the blob signal used for image-based reward computation.

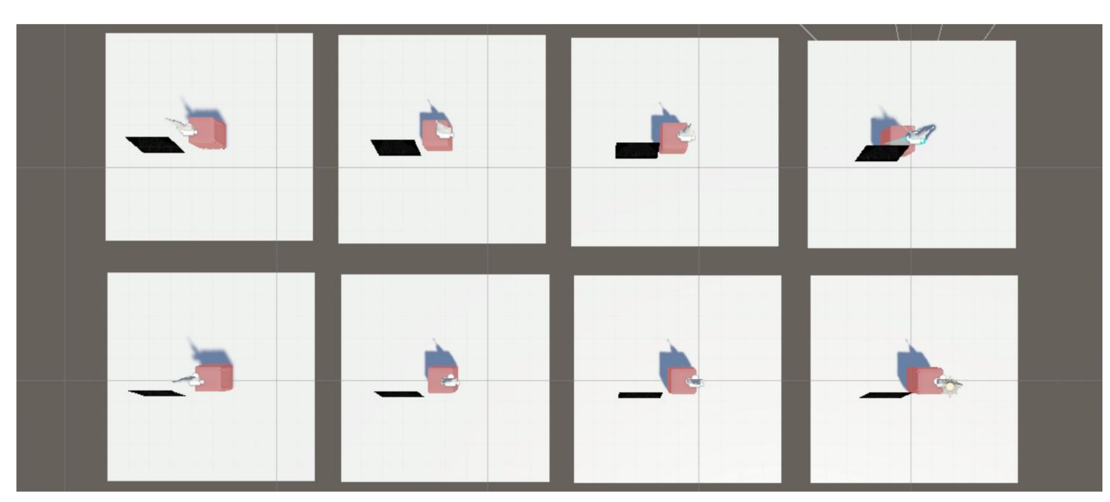

Figure 5: Eight parallel training environments running simultaneously in Unity to accelerate policy learning.

Results – Ablation Studies

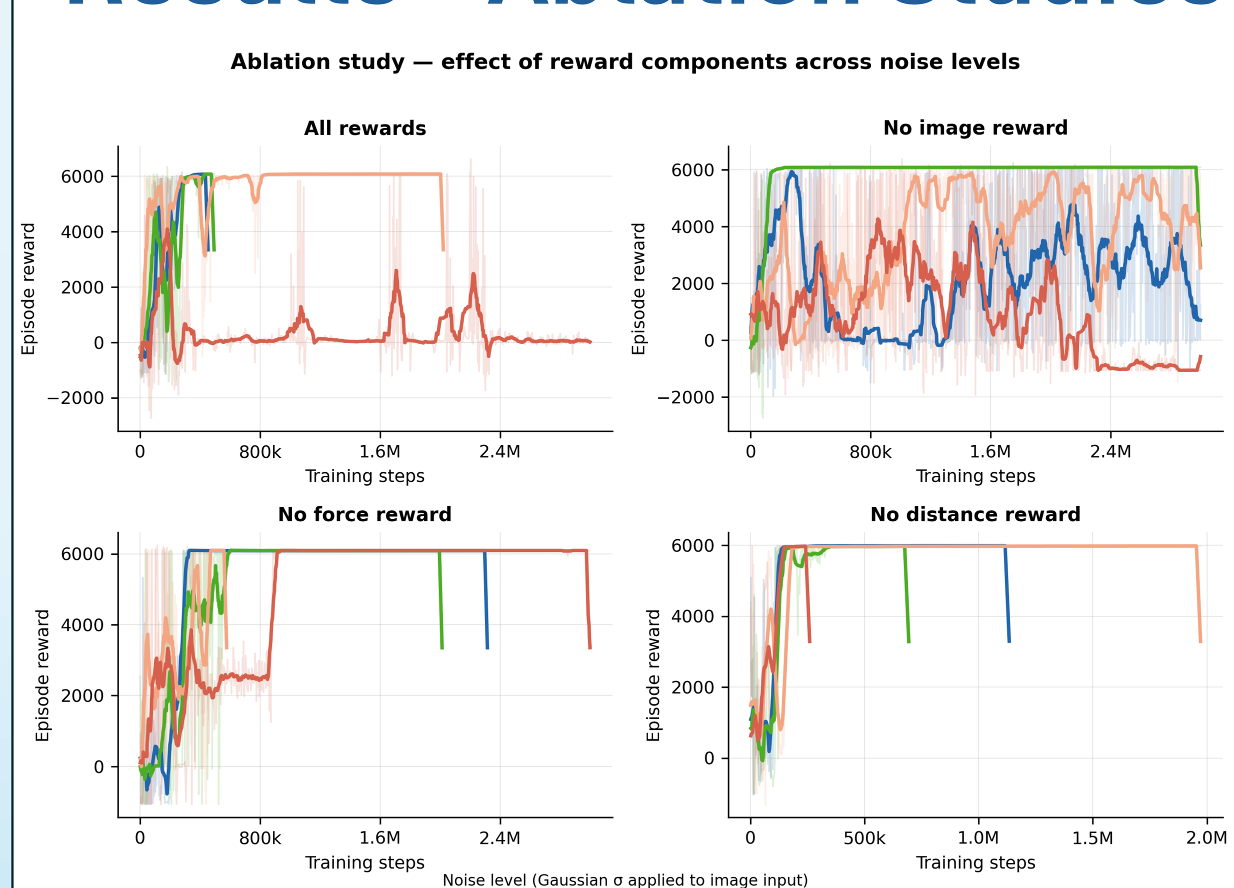

Figure 7: Training reward curves across all four ablation conditions and noise levels (σ = 0 to 0.015). Removing the image reward is the only condition that causes consistent training failure; no-force and no-distance conditions converge across all noise levels.

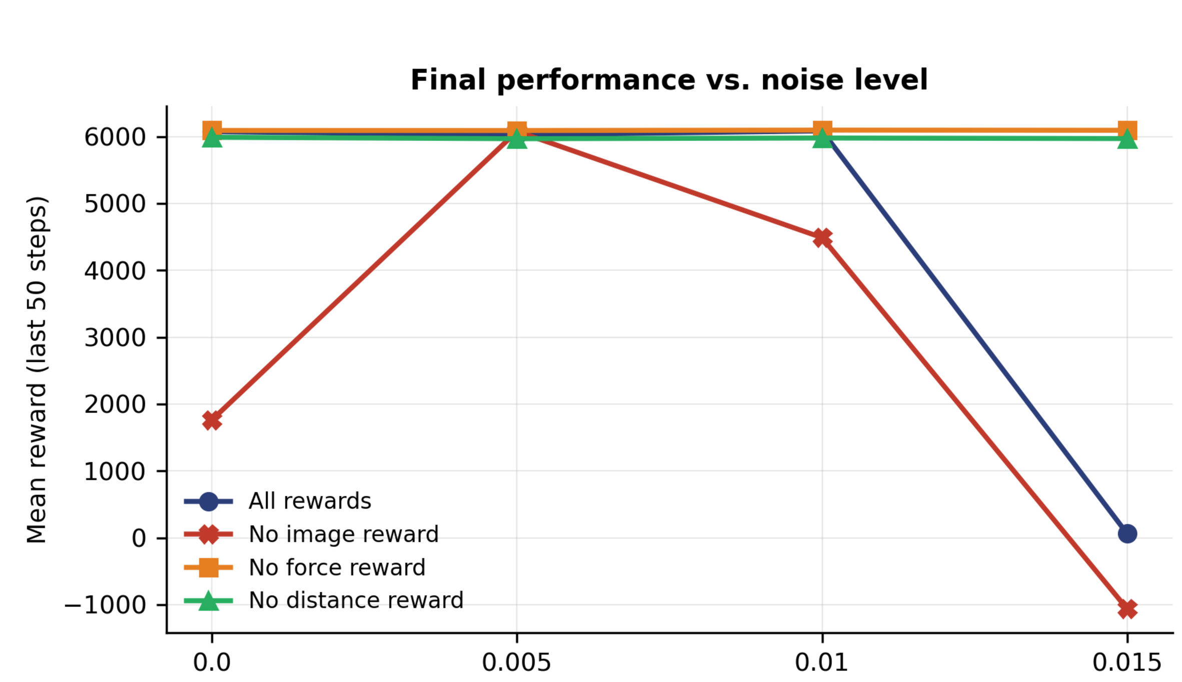

Figure 8: Final mean episodic reward (averaged over last 50 logged steps) as a function of noise level. The no-image-reward condition degrades sharply with noise, dropping below zero at σ = 0.015.

Discussion

What happens when the image reward is removed?

- Without image feedback, the agent cannot detect whether the tumor is visible in the ultrasound frame
- Training fails to converge reliably at most noise levels, confirming image reward as the most critical component
- The image reward is the primary signal connecting probe position to task success; distance and force alone are insufficient.

What happens when the force reward is removed?

- The agent still converges across all noise levels, reaching comparable end rewards
- Force regulation, preventing over-pressing and maintaining contact, appears to emerge implicitly from the image and distance rewards once the probe reaches the correct region.
- Force reward improves safety and tissue protection but is not load-bearing for training convergence

What happens when the distance reward is removed?

- Similar to no-force: the agent converges robustly across all noise levels.
- The image reward provides sufficient signal to guide the probe toward the target, making explicit distance shaping partially redundant once image feedback is available.
- Distance reward accelerates early training but is not critical for final performance.

Conclusions

- **The image reward is the most critical component.** Removing it causes training failure at most noise levels, as image noise directly corrupts the tumor detection signal the agent depends on.
- Force and distance rewards are robust to noise, so removing either does not prevent convergence across any tested condition.
- After training, the agent can rely on image and force cues alone, which is essential when the true target position is unknown.

Future Work

- Set up the robot physically and perform the zero-shot transfer

References

[1] Y. Ou and M. Tavakoli, “Learning Autonomous Surgical Irrigation and Suction with the da Vinci Research Kit Using Reinforcement Learning,” arXiv (Cornell University), Nov. 2024, doi: https://doi.org/10.48550/arxiv.2411.14622.

[2] Y. Ou and M. Tavakoli, “CRESSim-MPM: A Material Point Method Library for Surgical Soft Body Simulation with Cutting and Suturing,” arXiv (Cornell University), Feb. 2025, doi: https://doi.org/10.48550/arxiv.2502.18437.

Acknowledgements

I would like to acknowledge the support and contributions of Megan Zhou, Dr. Mahdi Tavakoli, and all members of the Telerobotic and Biorobotic Systems Group